Zhengyuan Lin 林政源

+81 70-7564-5942 | zylin@g.ecc.u-tokyo.ac.jp | zylinh.github.io

EDUCATION

The University of Tokyo Tokyo, Japan
M.S. in Information Science and Technology Apr. 2025 – Apr. 2027 (expected)

South China Normal University Guangzhou, China
B.E. in Computer Science and Technology Sep. 2020 – Jul. 2024

WORK EXPERIENCE

AI Intern in Alaya Lab Feb. 2026 – Present

- Contributes to research on interactive video world models and 3D/4D generation, with responsibilities spanning benchmark construction and model application.

Research Assistant in The Hong Kong Polytechnic University, DigiEnergy Lab Oct. 2024 – Apr. 2025

- Investigated and conducted research related to 2D/3D segmentation, Generative AI, and Embodied AI for renewable energy field.

RESEARCH EXPERIENCE

- **UT-Vision Lab, The University of Tokyo** *VLMs · Social Interaction Understanding*

  1. **Social Interaction Understanding via Nonverbal Cues** Oct. 2025 – Present

     Role: Research Lead

     - Investigates how vision-language models interpret nonverbal cues to infer human intent and anticipate future actions in social interactions.
     - Identified attention misalignment in VLMs' spatial and semantic gaze understanding and proposed attention-guided fine-tuning that aligns attention with gaze targets without adding extra modules.
     - Collects a multi-view, multi-person social interaction dataset whose complementary viewpoints reveal cues a single view misses, to better evaluate and improve VLMs' understanding of nonverbal cues and mental states.

- **Alaya Lab** *Interactive Video World Models · 3D Animation Generation*

  1. **WorldRover: Synthetic Video Data for World Exploration** May 2026 – Aug. 2026

     Role: Research Contributor; co-first author of [P1]

     - Contributed to a scalable UE-based video dataset collection pipeline that renders long-range explorations from first-person, third-person, and panoramic views with rich geometry, motion, camera, and action annotations.
     - Standardized heterogeneous Unreal Engine scenes, planned navigation trajectories, and validated assets for reliable headless rendering.

  2. **WorldMark: Action and Memory Benchmark for Interactive Video World Models** Feb. 2026 – Apr. 2026

     Role: Research Contributor

     - Constructed benchmark cases across visual styles and viewpoints, and conducted and visualized model evaluations under shared scenes and action sequences.
     - Contributed to a unified action interface mapped onto each model's native controls, a standardized test suite, and metrics that go beyond visual quality to diagnose per-axis action response and world memory.

  3. **AlayaWorld: Interactive Long-Horizon World Modeling** Feb. 2026 – Jul. 2026

     Role: Research Contributor

     - Contributed to an interactive video world model that generates controllable long-horizon environments under camera trajectories and dynamically switchable text prompts while preserving spatiotemporal consistency.

- Built and curated a prompt-conditioned dataset for magic event generation, covering combat, spell casting, and monster summoning.

4. Agentic Text-Guided Animation for Rigged 3D Assets Sep. 2026 – Present

Role: Research Lead

- Studies how models hold up as skeletal and motion complexity grow in text-guided animation for diverse rigged 3D assets, by constructing a difficulty-graded dataset of rigged assets and text prompts.
- Builds an agentic workflow that reads the rig, plans the motion, builds editable keyframe animation with tools, and iteratively refines it by rendering and inspecting its output.

- **DigiEnergy Lab, The Hong Kong Polytechnic University** *Generative AI · VLN*

1. Intention-Driven UAV Vision-Language Navigation Mar. 2025 – Present

Role: Research Lead

- Develops a city-scale UAV navigation benchmark and framework that replaces explicit landmark instructions with abstract human demands, grounding them into semantic destination goals and executable flight plans.
- Built a Hong Kong city knowledge base from OpenStreetMap and government open data, with diverse implicit demand expressions mapped to destinations.

2. Generative AI for PV Segmentation and Data Augmentation Jan. 2023 – Apr. 2025

Role: Research Lead; first author of [P4]

- Developed Stable Diffusion-based pipelines for generating paired synthetic photovoltaic panel (PV) imagery and segmentation masks to augment training data.
- Led method design and implementation across text-to-image synthesis, image-to-image translation, text-guided inpainting, and cross-attention-based annotation generation.

PUBLICATIONS

[P1] Xu, X.*, **Lin, Z.***, Li, R., Liu, Y., Zhang, K., & Ge, Y. (2026). WorldRover: A Scalable Synthetic Video Data Engine for World Exploration with Rich Annotations. *arXiv preprint*. arXiv:2608.15659. (* Equal contribution)

[P2] Xu, X., **Lin, Z.**, He, K., Feng, Y., Mao, X., Yin, Y., Ge, Y., & Zhang, K. (2026). WorldMark: A Unified Benchmark Suite for Interactive Video World Models. *arXiv preprint*. arXiv:2604.21686.

[P3] AlayaWorld Team (alphabetical; including **Lin, Z.**). (2026). AlayaWorld: Interactive Long-Horizon World Modeling—Full Technical Report. *arXiv preprint*. arXiv:2607.18367; v1.1: arXiv:2608.13492.

[P4] **Lin, Z.**, Guo, Z., Huang, D., Song, C., Tan, H., Song, X., Zhang, H., & Yan, J. (2023). Leveraging generative AI for renewable energy: Photovoltaic panel semantic segmentation case study. *Energy Proceedings*, 36. doi:10.46855/energy-proceedings-11182.

[P5] Tan, H., Guo, Z., **Lin, Z.**, Chen, Y., Huang, D., Yuan, W., Zhang, H., & Yan, J. (2024). General generative AI-based image augmentation method for robust rooftop PV segmentation. *Applied Energy* (IF: 12.2), 368, 123554. doi:10.1016/j.apenergy.2024.123554.

SKILLS AND LANGUAGES

Tools: PyTorch; Unreal Engine; CesiumJS

Languages: Mandarin (native); English (fluent); Japanese (fluent)